

①

Homogenous equation সোজা উদাহরণ

① $(x^1 + y^1)dx - y^1 dy = 0$ [power same]

② $(x^2 + x^1 y^1)dx - x^1 y^1 dy = 0$ [গুণ শাকল যোগ করে, power check]

③ $(y^1 + \sqrt{x^2 + y^2})dx - x^1 dy = 0 \rightarrow$ [power same]

$\downarrow$
 $(x^2 + y^2)^{1/2}$
 $x^2 x^{1/2} + y^2 x^{1/2}$
 $x^1 + y^1$

④ $v^3 dv - (v^3 - v^1)dv = 0$ = power same

Step-1
 $\rightarrow \frac{dy}{dx}$ form আনা, eqn ①

Step-2
 $v = \frac{y}{x}$
 $\Delta y = vx$
 Differentiating with x,
 $\Delta \frac{dy}{dx} = v \cdot \frac{dx}{dx} + x \frac{dv}{dx}$
 $\rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

Step 4
 $\rightarrow$ integration করা,
 $\rightarrow v$ এক মান বসানো
 $\rightarrow c$ বের করা eqn ②

Step 5
 $\therefore y(1) = 6$
 $\Rightarrow y(x) = y$
 x, y দেয়া থাকলে
 c এক মান বের করে
 eqn ② ত
 বসানো।

Step 3 : নতুন eqn থেকে dv সহ $v = \frac{y}{x}$ সহ x
 form বানানো

① $(y + \sqrt{x^2 + y^2}) dx - x dy = 0, y(1) = 0$ $\left[\int \frac{1}{\sqrt{a^2 + x^2}} = \ln [x + \sqrt{a^2 + x^2}] \right]$

$\therefore y + \sqrt{x^2 + y^2} dx = x dy$

$\Rightarrow \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \rightarrow \textcircled{1}$

$V = \frac{y}{x}$

$\Rightarrow y = Vx$

$\Rightarrow \frac{dy}{dx} = V \cdot 1 + x \frac{dV}{dx}$ put y & $\frac{dy}{dx}$ in eqn ①

$\therefore V + x \frac{dV}{dx} = \frac{Vx + \sqrt{x^2 + V^2 x^2}}{x}$

$\Rightarrow V = \frac{Vx + \sqrt{x^2(1+V^2)}}{x}$

$\Rightarrow V = \frac{Vx + x\sqrt{1+V^2}}{x}$

$\Rightarrow V = \frac{x(V + \sqrt{1+V^2})}{x}$

$\Rightarrow x \frac{dV}{dx} = V + \sqrt{1+V^2} - V$

$\Rightarrow x \frac{dV}{dx} = \sqrt{1+V^2}$

$\Rightarrow x dV = dx \sqrt{1+V^2}$

$\Rightarrow \frac{dV}{\sqrt{1+V^2}} = dx \frac{1}{x}$

$\Rightarrow \int \frac{1}{\sqrt{1+V^2}} dV = \int \frac{1}{x} dx = 0$

$\ln [V + \sqrt{1+V^2}] - \ln x = \ln c$

$\Rightarrow \ln \left[\frac{V + \sqrt{1+V^2}}{x} \right] = \ln c$

$\Rightarrow \frac{\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}}}{x} = c$

$\Rightarrow \frac{\frac{y}{x} + \sqrt{\frac{x^2 + y^2}{x^2}}}{x} = c$

$\Rightarrow \frac{\frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x}}{x} = c$

$\Rightarrow \frac{y + \sqrt{x^2 + y^2}}{x^2} = c$

$\Rightarrow \frac{y + \sqrt{x^2 + y^2}}{x^2} = c \rightarrow \textcircled{2}$

Given, $x=1, y=0$ put in eqn ②

$\therefore \frac{0 + \sqrt{1^2 + 0^2}}{1^2} = c$

$\Rightarrow c = 1$

$\therefore y + \sqrt{x^2 + y^2} = x^2$ (Ans)

$$(2) (x^2 - 3y^2) dx + 2xy dy = 0$$

$$\Rightarrow (x^2 - 3y^2) dx = -2xy dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 - 3y^2}{-2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{3y^2 - x^2}{2xy} \rightarrow (1)$$

$$V = \frac{y}{x}$$

$$\Rightarrow y = Vx$$

$$\Rightarrow \frac{dy}{dx} = V + x \frac{dV}{dx} \text{ in eqn (1)}$$

$$\Rightarrow V + x \frac{dV}{dx} = \frac{3V^2 x^2 - x^2}{2x^2 V}$$

$$\Rightarrow V + x \frac{dV}{dx} = \frac{x^2(3V^2 - 1)}{2x^2 V}$$

$$\Rightarrow x \frac{dV}{dx} = \frac{3V^2 - 1}{2V} - V$$

$$\Rightarrow x \frac{dV}{dx} = \frac{3V^2 - 1 - 2V^2}{2V}$$

$$\Rightarrow x \frac{dV}{dx} = \frac{V^2 - 1}{2V}$$

$$\Rightarrow \frac{dV}{\frac{V^2 - 1}{2V}} = \frac{dx}{x} \cdot \frac{1}{x}$$

$$\Rightarrow \int \frac{2V}{V^2 - 1} dV - \int \frac{1}{x} dx = 0$$

$$\left[\int \frac{f'(x)}{f(x)} = \ln[f(x)] \right]$$

$$\rightarrow \ln[V^2 - 1] - \ln[x] = \ln C$$

$$\Rightarrow \ln\left[\frac{V^2 - 1}{x}\right] = \ln C$$

$$\Rightarrow \frac{V^2 - 1}{x} = C \rightarrow (2)$$

$$\therefore \frac{\frac{y^2}{x^2} - 1}{x} = C$$

$$\Rightarrow \frac{y^2 - x^2}{x^3} = C$$

$$\Rightarrow \boxed{\frac{y^2 - x^2}{x^3} = C} \text{ (Ans)}$$

$$(3) v^3 du - (u^3 - uv^2) dv = 0$$

$$\frac{du}{dv} = \frac{u^3 - uv^2}{v^3}$$

$$z = \frac{u}{v} \Rightarrow u = zv$$

$$\frac{du}{dv} = z + v \frac{dz}{dv}$$

$$z + v \frac{dz}{dv} = \frac{z^3 v^3 - z v^3}{v^3}$$

$$v \frac{dz}{dv} = z^3 - z - 2$$

$$\frac{1}{z^3 - 2z} dz + \frac{1}{v} dv = 0$$

$$\int \frac{1}{z(z^2 - 2)} dz + \int \frac{1}{v} dv = 0$$

Partial;

$$\frac{1}{z(z+\sqrt{2})(z-\sqrt{2})} = \frac{A}{z} + \frac{B}{z+\sqrt{2}} + \frac{C}{z-\sqrt{2}}$$

$$1 = A(z+\sqrt{2})(z-\sqrt{2}) + B\{z(z-\sqrt{2})\} + C\{z(z+\sqrt{2})\}$$

$$1 = A(z^2 - 2) + B(z^2 - z\sqrt{2}) + C(z^2 + z\sqrt{2})$$

$$z=0 \Rightarrow 1 = A(0-2) + 0 + 0$$

$$\Rightarrow A = -\frac{1}{2}$$

$$z = +\sqrt{2}$$

$$\Rightarrow 1 = 0 + 0 + 4C$$

$$\Rightarrow C = \frac{1}{4}$$

$$z = -\sqrt{2}$$

$$\Rightarrow 1 = 0 + 4B + 0$$

$$\Rightarrow B = \frac{1}{4}$$

$$\int \left(-\frac{1}{2z} + \frac{1}{4(z+\sqrt{2})} + \frac{1}{4(z-\sqrt{2})} \right) dz + \int \frac{1}{v} dv = 0$$

$$-\frac{1}{2} \int \frac{1}{z} dz + \frac{1}{4} \int \frac{1}{z+\sqrt{2}} dz + \frac{1}{4} \int \frac{1}{z-\sqrt{2}} dz$$

$$\Rightarrow -\frac{1}{2} \ln(z) + \frac{1}{4} \ln(z+\sqrt{2}) + \frac{1}{4} \ln(z-\sqrt{2})$$

$$(-\frac{1}{2} \ln(z) + \frac{1}{4} \ln(z^2 - 2) - \ln(v) = 0) \times 4$$

$$\Rightarrow -2 \ln(z) + \ln(z^2 - 2) - 4 \ln(v) = 0$$

$$\Rightarrow -\ln(z)^2 + \ln(z^2 - 2) - \ln(v)^4 = \ln(1)$$

$$\Rightarrow \frac{z^2 - 2}{z^2 v^4} = C$$

$$\Rightarrow \frac{u^2}{v^2} = 2$$

$$\frac{u^2}{v^2} \times v^4 = 2C$$

$$\Rightarrow \frac{u^2 - 2v^2}{u^2 v^2} = C$$

$$\Rightarrow u^2 - 2v^2 = C u^2 v^4$$

$$(4) (2xy + 3y^2) dx - (2xy + x^2) dy = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy + 3y^2}{2xy + x^2} \rightarrow (1)$$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx} \quad \because y = vx$$

$$\therefore v + x \frac{dv}{dx} = \frac{2x^2v + 3v^2x^2}{2x^2v + x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^2(2v + 3v^2)}{x^2(2v + 1)}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v + 3v^2}{2v + 1} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v + 3v^2 - 2v^2 - v}{2v + 1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 + v}{2v + 1}$$

$$\Rightarrow \frac{dv}{\frac{v^2 + v}{2v + 1}} = dx \frac{1}{x}$$

$$\Rightarrow \int \frac{2v + 1}{v^2 + v} dv - \int \frac{1}{x} dx = 0$$

$$\Rightarrow \ln[v^2 + v] - \ln[x] = \ln c$$

$$\Rightarrow \ln\left[\frac{v^2 + v}{x}\right] = \ln c$$

$$\Rightarrow \frac{v^2 + v}{x} = c$$

$$\Rightarrow \frac{\frac{y^2}{x^2} + \frac{y}{x}}{x} = c$$

$$\Rightarrow \frac{y^2 + xy}{x^3} = c$$

$$\Rightarrow \frac{y^2 + xy}{x^3} = c \quad (Ans)$$

$$(5) (x^2 + 3y^2) dx - 2xy dy = 0, y(2) = 6$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy} \rightarrow (1)$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx} \quad \because y = vx$$

$$\therefore v + x \frac{dv}{dx} = \frac{x^2 + 3x^2v^2}{2vx^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^2(1 + 3v^2)}{x^2 2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + 3v^2}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + 3v^2 - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$\Rightarrow dv \frac{2v}{1 + v^2} = dx \frac{1}{x}$$

$$\Rightarrow \int \frac{2v}{1 + v^2} dv - \int \frac{1}{x} dx = 0 \quad \left[\int \frac{f'(x)}{f(x)} = \ln f(x) \right]$$

$$\Rightarrow \ln(1 + v^2) - \ln(x) = \ln(c)$$

$$\Rightarrow \ln\left(\frac{1 + v^2}{x}\right) = \ln c$$

$$\Rightarrow \frac{1 + \frac{y^2}{x^2}}{x} = c$$

$$\Rightarrow \frac{x^2 + y^2}{x^3} = c$$

$$\Rightarrow \frac{x^2 + y^2}{x^3} = c$$

$$\therefore y(2) = 6$$

$$\therefore \frac{4 + 36}{8} = c$$

$$\Rightarrow c = 5$$

$$\Rightarrow x^2 + y^2 = 5x^3 \quad (Ans)$$

(6) $(2x-5y)dx + (4x-y)dy = 0, y(1) = 4$

$\Rightarrow \frac{dy}{dx} = \frac{-(2x-5y)}{4x-y}$

$\Rightarrow \frac{dy}{dx} = \frac{5y+2x}{4x-y}$

$v = y/x$

$y = vx$

$\frac{dy}{dx} = v + x \frac{dv}{dx}$

$\therefore v + x \frac{dv}{dx} = \frac{5vx+2x}{4x-vx}$

$\Rightarrow v + x \frac{dv}{dx} = \frac{5v+2}{4-v}$

$\Rightarrow x \frac{dv}{dx} = \frac{5v+2}{4-v} - v$

$\Rightarrow x \frac{dv}{dx} = \frac{5v+2-(4v-v^2)}{4-v}$

$\Rightarrow x \frac{dv}{dx} = \frac{5v+2-4v+v^2}{4-v}$

$\Rightarrow x \frac{dv}{dx} = \frac{v^2+v+2}{4-v}$

$\Rightarrow dv \frac{4-v}{v^2+v+2} = dx \frac{1}{x}$

$\Rightarrow \int \frac{4-v}{v^2+v+2} dv = \int \frac{1}{x} dx = 0$

Partial

$\frac{4-v}{(v+2)(v-1)} = \frac{A}{v+2} + \frac{B}{v-1}$

$\Rightarrow 4-v = A(v-1) + B(v+2)$

$\therefore A-1=0$

$\Rightarrow v=1$

$\therefore 4-1 = A(1-1) + B(1+2)$

$\Rightarrow 3=3B$

$\Rightarrow B=1$

$\therefore v+2=0$

$\Rightarrow v=-2$

$\therefore 4-(-2) = A(-2-1) + 0$

$\Rightarrow 6 = -3A$

$\Rightarrow A = -2$

$\int \frac{4-v}{(v+2)(v-1)} dv = \int \left(-\frac{2}{v+2} + \frac{1}{v-1} \right) dv$

$= -2 \ln(v+2) + \ln(v-1)$

$= \ln(v+2)^{-2} + \ln(v-1)$

$= \ln \left\{ \frac{v-1}{(v+2)^2} \right\}$

$\Rightarrow \ln \left\{ \frac{v-1}{(v+2)^2} \right\} - \ln x = \ln c$

$\Rightarrow \frac{v-1}{x(v+2)^2} = c$

$\Rightarrow \frac{\frac{y}{x}-1}{x \left(\frac{y}{x}+2 \right)^2} = c$

$\Rightarrow \frac{\frac{y-x}{x}}{x \left(\frac{y}{x}+2 \right)^2} = c$

$\Rightarrow \frac{\frac{y-x}{x}}{x \left(\frac{y+2x}{x} \right)^2} = c$

$\Rightarrow \frac{\frac{y-x}{x}}{\frac{(y+2x)^2}{x}} = c$

$y(1) = 4$

$\therefore \frac{4-1}{(4+2)^2} = c$

$\Rightarrow \frac{3}{36} = c$

$\Rightarrow \frac{1}{12} = c$

$\Rightarrow \frac{y-x}{(y+2x)^2} = \frac{1}{12}$

$\Rightarrow \frac{y-x}{(y+2x)^2} = c$

(12)